M18-01 — Preorder traversal

Root → left subtree → right subtree

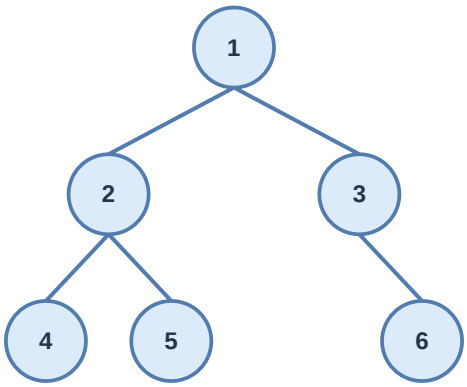

Diagram note

Preorder: 1 2 4 5 3 6

M18-02 — Inorder and postorder

Two traversal orders of the same tree

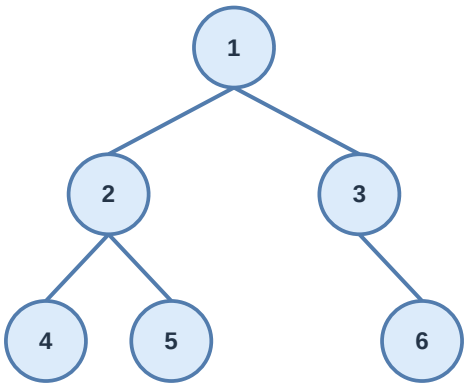

Diagram note

Inorder: 4 2 5 1 3 6 | Postorder: 4 5 2 6 3 1

M18-03 — Count nodes and leaves

Leaves have no children

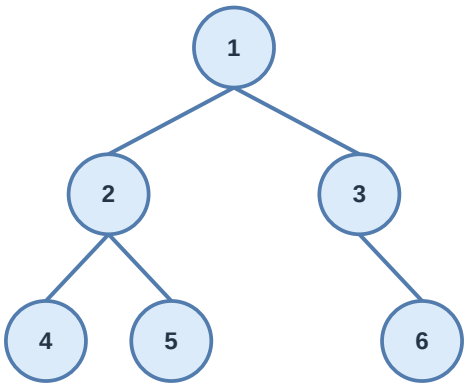

Diagram note

6 actual nodes; 3 leaves: 4, 5, 6

M18-04 — BST search

Search path for target 6

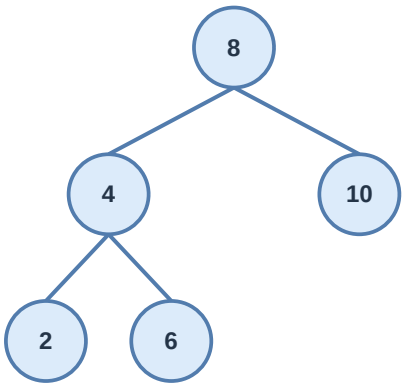

Diagram note

Search path: 8 → 4 → 6

M18-05 — BST minimum and maximum

Leftmost = minimum; rightmost = maximum

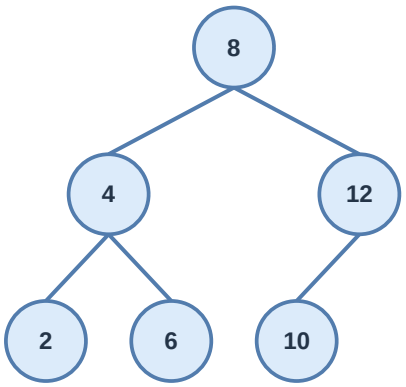

Diagram note

Minimum = 2; maximum = 12

M18-06 — Kth smallest

Inorder values are in ascending order

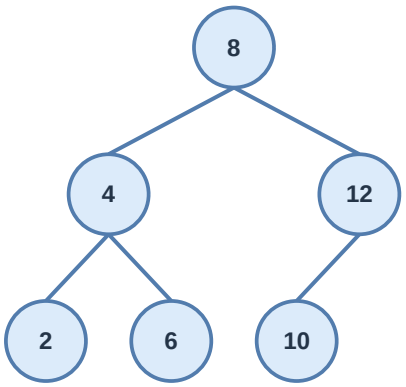

Diagram note

Inorder: 2 4 6 8 10 12; K = 3 gives 6

M18-07 — Tree height

Height is measured in nodes

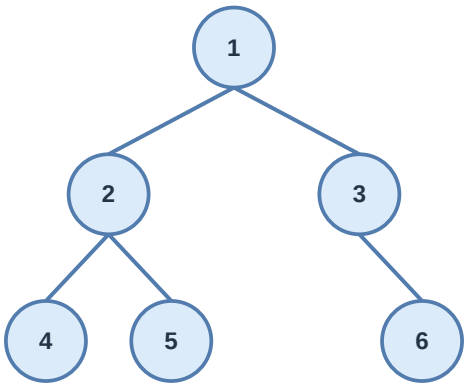

Diagram note

Longest root-to-leaf path has 3 nodes

M18-08 — Identical trees

Compare corresponding values and structure

Diagram note

Tree A has right child 3; tree B has right child 4. Result: NO

M18-09 — Level-order traversal

Breadth-first, one level per line

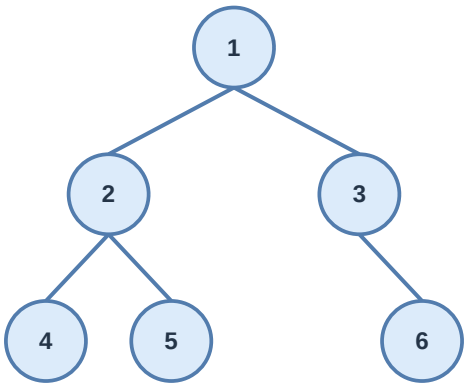

Diagram note

Levels: [1] / [2, 3] / [4, 5, 6]

M18-10 — Validate BST

A node must satisfy all ancestor bounds

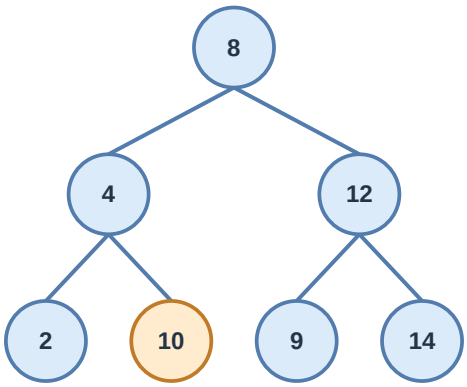

Diagram note

Node 10 is inside the left subtree of 8 but 10 > 8: INVALID

M18-11 — Lowest common ancestor

LCA of 2 and 6 is 4

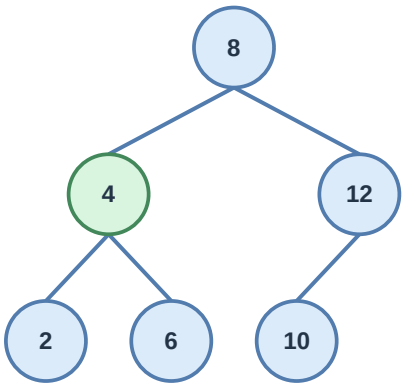

Diagram note

LCA = 4

M18-12 — Delete from BST

Replace two-child node 4 with successor 6

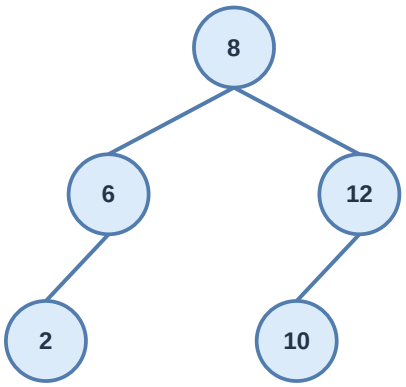

Diagram note

Preorder after deletion: 8 6 2 12 10

M18-13 — Bottom-up max heap

Bottom-up construction result for 1 2 3 4 5

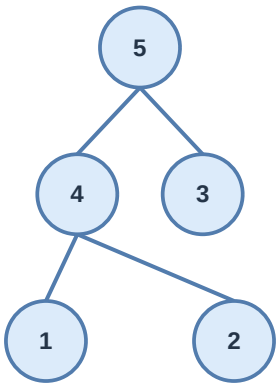

Diagram note

Heap array: 5 4 3 1 2

M18-14 — Running median

Two heaps partition the lower and upper halves

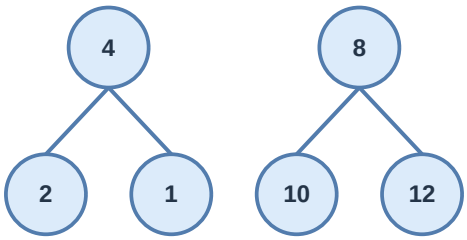

Diagram note

Illustrative partition only; sample stream medians are 5.0, 3.5, 5.0, 4.5, 5.0

M18-15 — Merge two BSTs

Merge both inorder sequences

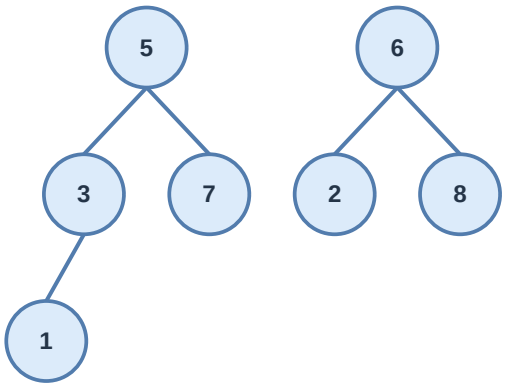

Diagram note

Merged order: 1 2 3 5 6 7 8

M18-16 — Serialize and deserialize

Preorder serialization marks each missing child with #

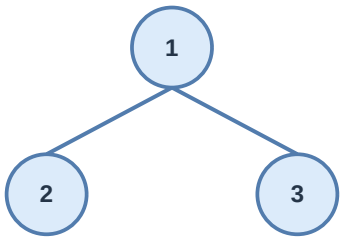

Diagram note

Serialization: 1 2 # # 3 # #

M18-17 — Tree diameter

Highlighted path contains four edges

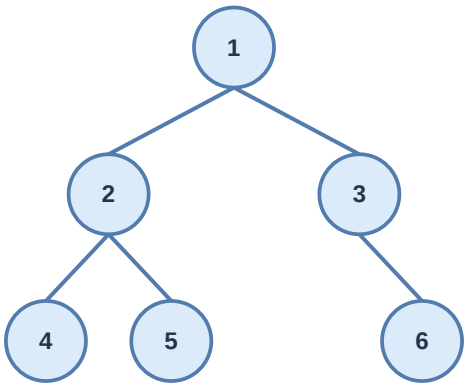

Diagram note

Diameter path: 4 → 2 → 1 → 3 → 6; diameter = 4

M18-18 — Recover swapped values

Swap the misplaced values 3 and 2

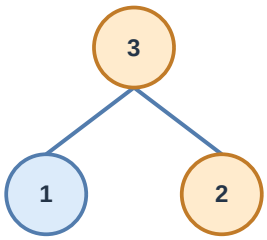

Diagram note

Corrected inorder: 1 2 3

M18-19 — Kth largest stream

Min heap stores the K largest values seen so far

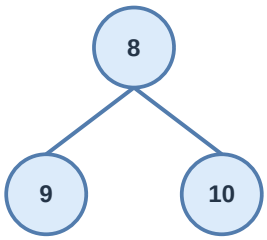

Diagram note

For K = 3, answers: NONE NONE 4 4 5 8

M18-20 — Maximum-sum BST subtree

Entire sample tree is a valid BST

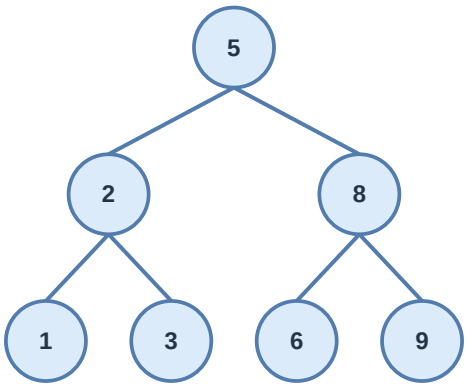

Diagram note

Sum = 34